

DATA & PICTOGRAPHS

Reading and Making Picture Graphs

Planrs Academy

Learning Objectives

- Understand what DATA means
- Learn the 4 parts of a PICTOGRAPH
- Read a pictograph to answer questions
- Create your own pictograph using real data

SECTION 1 — WHAT IS DATA?

What is Data?

DATA = Information we collect about something!

Examples of Data Around Us

★	A teacher marking attendance in a register
★	A shopkeeper counting vegetables at the sabzi mandi
★	A child counting rotis made for dinner
★	Survey results from the whole class
★	Weather information (sunny, rainy, cloudy)

How Do We Collect Data?

We collect data by **asking questions**. Here are some examples:

Q1	How many students walk to school?
Q2	What is your favourite festival food?
Q3	Which fruit did you bring in your tiffin today?
Q4	How many students were absent this week?
Q5	What sport do you like best?

Organising Data into Categories

After collecting data we sort it into **CATEGORIES** — similar groups:

Category Group	Examples
Languages	Hindi, Tamil, Telugu, Marathi, Bengali
Diwali Sweets	Ladoo, Barfi, Chakli, Jalebi

Transport to School	Auto-rickshaw, Bus, Walking, Car, Cycle
Farm Animals	Cow, Goat, Buffalo, Hen, Dog

SECTION 2 — WHAT IS A PICTOGRAPH?

What is a Pictograph?

A **PICTOGRAPH** (also called a Picture Graph) uses **pictures or symbols** to show data. Instead of writing numbers, we draw pictures. Each picture stands for a **certain number of things**.

The 4 Key Parts of a Pictograph

1	TITLE	Tells us what the pictograph is about.
2	CATEGORIES	The groups or items we are comparing.
3	PICTURES	Symbols drawn — each one stands for a number.
4	KEY	Explains what one picture / symbol stands for.

Steps to Build a Pictograph

Step 1	Add TITLE — Write what your pictograph is about at the top.
Step 2	Add CATEGORIES — List the groups you are comparing down the left side.
Step 3	Add PICTURES — Draw the correct number of symbols in each row.
Step 4	Add KEY — State what one symbol means, e.g. ★ = 1 student.

SECTION 3 — PICTOGRAPH EXAMPLES

Example 1: Favourite Fruits of Class 2A

Class 2A voted for their favourite fruit. Here is the data shown as a pictograph where each star (★) represents 1 student.

Category	Pictograph (★ = 1)	Total
Mango	★ ★ ★ ★ ★	5
Banana	★ ★ ★	3
Apple	★ ★ ★ ★	4
Guava	★ ★	2
KEY: ★ = 1 student		

Questions about the Fruits Pictograph

No.	Question	Answer
Q1	Which fruit is liked the MOST?	Mango — 5 students
Q2	Which fruit is liked the LEAST?	Guava — 2 students
Q3	How many students like Apple?	4 students
Q4	How many MORE students like Mango than Guava?	5 – 2 = 3 more
Q5	How many students voted in total?	5+3+4+2 = 14

Example 2: How We Come to School

Students told their teacher how they travel to school each day. Each house symbol (■) stands for 1 student.

Category	Pictograph (■ = 1)	Total
Auto-rickshaw	■ ■ ■ ■ ■ ■	6
School Bus	■ ■ ■ ■	4
Walking	■ ■ ■ ■ ■	5
Car	■ ■ ■	3
Cycle	■ ■	2
KEY: ■ = 1 student		

Questions about the Transport Pictograph

No.	Question	Answer
Q1	How many students come by auto-rickshaw?	6 students
Q2	Which is the LEAST popular way to travel?	Cycle — only 2
Q3	Do more students walk or come by car?	More walk (5 > 3)
Q4	How many students in total?	6+4+5+3+2 = 20
Q5	How many more come by auto than bus?	6 – 4 = 2 more

SECTION 3 CONTINUED — MORE EXAMPLES

Example 3: Favourite Diwali Sweets

The teacher asked the class: "Which Diwali sweet is your favourite?" Each circle (●) stands for 1 student.

Category	Pictograph (● = 1)	Total
Ladoo	● ● ● ● ● ● ●	7
Barfi	● ● ● ● ●	5
Jalebi	● ● ● ●	4
Chakli	● ● ●	3
KEY: ● = 1 student		

Questions about the Diwali Sweets Pictograph

No.	Question	Answer
Q1	Which sweet is the most popular?	Ladoo — 7 students
Q2	Which sweet has the least votes?	Chakli — 3 students
Q3	How many more like Ladoo than Jalebi?	7 – 4 = 3 more
Q4	How many students are in the class altogether?	7+5+4+3 = 19

SECTION 4 — TIPS, TRICKS & FUN FACTS

Did You Know?

★	PICTOGRAPH = PICTO (picture) + GRAPH (chart) = a PICTURE CHART!
★	In India, election results on TV often use bar graphs and pictographs.
★	Weather reports use picture symbols for sunny, rainy, and cloudy days.
★	Graphs are used in cricket scorecards, report cards, and census data!

Memory Checklist

Every time you read a pictograph, check these 3 things:

✓ TITLE	What is the pictograph about?
✓ PICTURES	How many pictures are in each row?
✓ KEY	What does one picture stand for?

SECTION 5 — LET'S MAKE ONE!

Activity: Create Your Own Pictograph

Use the data below to draw your own pictograph. Follow all 4 steps and write the KEY!

Data to use:

Diwali Sweet	Number of Students
Ladoo	5 students
Barfi	3 students
Jalebi	4 students
Chakli	2 students

Draw your pictograph below:

TITLE: _____

Sweet	Draw your pictures here (use ★ for each student)	Count
Ladoo		
Barfi		
Jalebi		
Chakli		

KEY:

Now answer these questions about your pictograph:

Which sweet is the most popular? → _____

Which sweet has the fewest votes? → _____

How many students voted altogether? → _____

How many MORE students like Ladoo than Chakli? → _____

SECTION 6 — PRACTICE QUESTIONS

Test Yourself!

Read the pictograph below, then answer all six questions.

Category	Pictograph (★ = 1)	Total
Cricket	★ ★ ★ ★ ★ ★ ★ ★	8
Football	★ ★ ★ ★ ★	5
Kabaddi	★ ★ ★	3
Badminton	★ ★ ★ ★ ★ ★	6
KEY: ★ = 1 student		

Write your answers below:

Q1	Which sport is the most popular in the class? Answer: _____
Q2	Which sport has the fewest students? Answer: _____
Q3	How many students like Badminton? Answer: _____
Q4	How many MORE students like Cricket than Kabaddi? Answer: _____
Q5	How many students were surveyed altogether? Answer: _____

Q6

Which two sports together equal the same count as Cricket?

Answer: _____

SECTION 7 — ANSWER KEY

Answer Key for Practice Questions

Check your answers — try all questions first before looking here!

No.	Question	Answer
Q1	Which sport is the most popular?	Cricket — 8 students
Q2	Which sport has the fewest students?	Kabaddi — 3 students
Q3	How many students like Badminton?	6 students
Q4	How many MORE like Cricket than Kabaddi?	$8 - 3 = 5$ more
Q5	How many students were surveyed altogether?	$8+5+3+6 = 22$
Q6	Which two sports together equal Cricket's count?	Football (5) + Kabaddi (3) = 8

What We Learned Today

- DATA is information we collect by counting or asking questions.
- A PICTOGRAPH uses pictures or symbols to display data visually.
- Every pictograph has 4 parts: TITLE, CATEGORIES, PICTURES, KEY.
- We can READ a pictograph to answer questions about most, least, and differences.
- We can MAKE our own pictograph using real survey data.

YOU ARE A DATA STAR!
Great work today, Class! — Planrs Academy